

Assignment -1

Linux Environment, Git Version Control & Branch Merging — Lab Submission

Question 1 — Linux Environment and File Management (5 Marks)

Created the project directory **Bhoomika-GLA-CourseRegistration** with subdirectories **src**, **docs**, **backup**, created **course_info.txt** with student/project details, and demonstrated: current working directory (pwd), directory contents (ls), file contents (cat), and the project directory structure (tree).

Creating the directory structure and course_info.txt:

```
PS C:\Users\DELL> cd C:\Users\DELL\Desktop\Bhoomika-GLA-CourseRegistration
PS C:\Users\DELL\Desktop\Bhoomika-GLA-CourseRegistration> pwd

Path
----
C:\Users\DELL\Desktop\Bhoomika-GLA-CourseReg...

PS C:\Users\DELL\Desktop\Bhoomika-GLA-CourseRegistration> ls
PS C:\Users\DELL\Desktop\Bhoomika-GLA-CourseRegistration> mkdir src, docs, backup

Directory: C:\Users\DELL\Desktop\Bhoomika-G
LA-CourseRegistration

Mode                LastWriteTime         Length
----                -
d-----          05-09-2026         23:40
d-----          05-09-2026         23:40
d-----          05-09-2026         23:40

PS C:\Users\DELL\Desktop\Bhoomika-GLA-CourseRegistration> @"
>> Student Name: Bhoomika
>> Roll Number: 2415001879
>> Subject Name: <your subject>
>> Project Name: Online Course Registration System
>> Semester: 5th Semester
>> "@ | Out-File -FilePath course_info.txt -Encoding utf8
PS C:\Users\DELL\Desktop\Bhoomika-GLA-CourseRegistration>
PS C:\Users\DELL\Desktop\Bhoomika-GLA-CourseRegistration> pwd

Path
----
C:\Users\DELL\Desktop\Bhoomika-GLA-CourseReg...

PS C:\Users\DELL\Desktop\Bhoomika-GLA-CourseRegistration> ls
```

Directory structure, file contents, and Git initialization begins:

Directory: C:\Users\DELL\Desktop\Bhoomika-G
LA-CourseRegistration

Mode		LastWriteTime	Length
----		-----	-----
d-----	05-09-2026	23:40	
d-----	05-09-2026	23:40	
d-----	05-09-2026	23:40	
-a----	05-09-2026	23:41	151

PS C:\Users\DELL\Desktop\Bhoomika-GLA-CourseRegistration> cat course_info.txt

Student Name: Bhoomika
Roll Number: 2415001879
Subject Name: <your subject>
Project Name: Online Course Registration System
Semester: 5th Semester

PS C:\Users\DELL\Desktop\Bhoomika-GLA-CourseRegistration> tree

Folder PATH listing for volume OS
Volume serial number is 0000002C F08F:9F96

C:.
├── backup
├── docs
└── src

PS C:\Users\DELL\Desktop\Bhoomika-GLA-CourseRegistration> pwd

Path

C:\Users\DELL\Desktop\Bhoomika-GLA-CourseReg...

PS C:\Users\DELL\Desktop\Bhoomika-GLA-CourseRegistration> git init

Initialized empty Git repository in C:/Users/DELL/Desktop/Bhoomika-GLA-CourseRegistration/.git/

PS C:\Users\DELL\Desktop\Bhoomika-GLA-CourseRegistration> git config user.name "Bhoomika"

PS C:\Users\DELL\Desktop\Bhoomika-GLA-CourseRegistration> git config user.email "your_email@example.com"

PS C:\Users\DELL\Desktop\Bhoomika-GLA-CourseRegistration> git status

On branch master

No commits yet

Question 2 — Git Repository and Version Control (6 Marks)

Initialized the Git repository, configured user identity, checked status, staged and committed all project files, then modified `course_info.txt` to add course/program name and a project description, viewed the diff, and committed the update.

git init, git config, git status (no commits yet):

```
Directory: C:\Users\DELL\Desktop\Bhoomika-G
LA-CourseRegistration

Mode                LastWriteTime         Length
----                -
d-----          05-09-2026    23:40
d-----          05-09-2026    23:40
d-----          05-09-2026    23:40
-a-----          05-09-2026    23:41             151

PS C:\Users\DELL\Desktop\Bhoomika-GLA-CourseRegistration> cat course_info.txt
Student Name: Bhoomika
Roll Number: 2415001879
Subject Name: <your subject>
Project Name: Online Course Registration System
Semester: 5th Semester
PS C:\Users\DELL\Desktop\Bhoomika-GLA-CourseRegistration> tree
Folder PATH listing for volume OS
Volume serial number is 0000002C F08F:9F96
C:.
├── backup
├── docs
└── src
PS C:\Users\DELL\Desktop\Bhoomika-GLA-CourseRegistration> pwd

Path
----
C:\Users\DELL\Desktop\Bhoomika-GLA-CourseReg...

PS C:\Users\DELL\Desktop\Bhoomika-GLA-CourseRegistration> git init
Initialized empty Git repository in C:/Users/DELL/Desktop/Bhoomika-GLA-CourseRegistration/.git/
PS C:\Users\DELL\Desktop\Bhoomika-GLA-CourseRegistration> git config user.name "Bhoomika"
PS C:\Users\DELL\Desktop\Bhoomika-GLA-CourseRegistration> git config user.email "your_email@example.com"
PS C:\Users\DELL\Desktop\Bhoomika-GLA-CourseRegistration> git status
On branch master

No commits yet
```

Staging, initial commit, file modification, status, and diff:

```

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        course_info.txt

nothing added to commit but untracked files present (use "git add" to track)
PS C:\Users\DELL\Desktop\Bhoomika-GLA-CourseRegistration> git add .
warning: in the working copy of 'course_info.txt', LF will be replaced by CRLF the next time Git touches it
PS C:\Users\DELL\Desktop\Bhoomika-GLA-CourseRegistration> git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   course_info.txt

PS C:\Users\DELL\Desktop\Bhoomika-GLA-CourseRegistration> git commit -m "Initial commit: project structure and course_info.txt"
[master (root-commit) 48ccae8] Initial commit: project structure and course_info.txt
1 file changed, 5 insertions(+)
 create mode 100644 course_info.txt
PS C:\Users\DELL\Desktop\Bhoomika-GLA-CourseRegistration> Add-Content course_info.txt "`nCourse/Program Name: B.Tech Computer Science`nProject Description:
A web-based system for managing student course registrations, allowing students to enroll in subjects and view their semester schedule."
PS C:\Users\DELL\Desktop\Bhoomika-GLA-CourseRegistration> cat course_info.txt
Student Name: Bhoomika
Roll Number: 2415001879
Subject Name: <your subject>
Project Name: Online Course Registration System
Semester: 5th Semester

Course/Program Name: B.Tech Computer Science
Project Description: A web-based system for managing student course registrations, allowing students to enroll in subjects and view their semester schedule.
PS C:\Users\DELL\Desktop\Bhoomika-GLA-CourseRegistration> git status
On branch master
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   course_info.txt

no changes added to commit (use "git add" and/or "git commit -a")
PS C:\Users\DELL\Desktop\Bhoomika-GLA-CourseRegistration> git diff
warning: in the working copy of 'course_info.txt', LF will be replaced by CRLF the next time Git touches it

```

Diff output and final commit of the modification:

```

diff --git a/course_info.txt b/course_info.txt
index 8396e76..692a6bf 100644
--- a/course_info.txt
+++ b/course_info.txt
@@ -3,3 +3,6 @@ Roll Number: 2415001879
 Subject Name: <your subject>
 Project Name: Online Course Registration System
 Semester: 5th Semester
+
+Course/Program Name: B.Tech Computer Science
+Project Description: A web-based system for managing student course registrations, allowing students to enroll in subjects and view their semester schedule
.
PS C:\Users\DELL\Desktop\Bhoomika-GLA-CourseRegistration> git add course_info.txt
warning: in the working copy of 'course_info.txt', LF will be replaced by CRLF the next time Git touches it
PS C:\Users\DELL\Desktop\Bhoomika-GLA-CourseRegistration> git commit -m "Updated course_info.txt with course/program name and project description"
[master b042622] Updated course_info.txt with course/program name and project description
1 file changed, 3 insertions(+)
PS C:\Users\DELL\Desktop\Bhoomika-GLA-CourseRegistration> git branch course-search
PS C:\Users\DELL\Desktop\Bhoomika-GLA-CourseRegistration> git checkout course-search
Switched to branch 'course-search'
PS C:\Users\DELL\Desktop\Bhoomika-GLA-CourseRegistration> Add-Content src\course_search.txt "Course Search Module`nThis module allows students to search for
available courses by name, subject, or semester. It provides filtering options and displays course details such as course code, credits, and instructor nam
e."
PS C:\Users\DELL\Desktop\Bhoomika-GLA-CourseRegistration> cat src\course_search.txt
Course Search Module
This module allows students to search for available courses by name, subject, or semester. It provides filtering options and displays course details such as
course code, credits, and instructor name.
PS C:\Users\DELL\Desktop\Bhoomika-GLA-CourseRegistration> git add src\course_search.txt
warning: in the working copy of 'src/course_search.txt', LF will be replaced by CRLF the next time Git touches it
PS C:\Users\DELL\Desktop\Bhoomika-GLA-CourseRegistration> git commit -m "Added course search module description"
[course-search 68bc0ce] Added course search module description
1 file changed, 2 insertions(+)
 create mode 100644 src/course_search.txt
PS C:\Users\DELL\Desktop\Bhoomika-GLA-CourseRegistration> git branch
* course-search
  master
PS C:\Users\DELL\Desktop\Bhoomika-GLA-CourseRegistration> git branch --show-current
course-search
PS C:\Users\DELL\Desktop\Bhoomika-GLA-CourseRegistration> git checkout master
Switched to branch 'master'
PS C:\Users\DELL\Desktop\Bhoomika-GLA-CourseRegistration> ls src

```

Question 3 — Feature Branch Development (7 Marks)

Created a new branch **course-search**, switched to it, added **course_search.txt** inside **src** describing the Course Search Module, staged and committed it, listed all branches, displayed the current branch, and switched back to master.

Branch creation, checkout, file creation, staging and commit:


```

diff --git a/course_info.txt b/course_info.txt
index 8396e76..692a6bf 100644
--- a/course_info.txt
+++ b/course_info.txt
@@ -3,3 +3,6 @@ Roll Number: 2415001879
Subject Name: <your subject>
Project Name: Online Course Registration System
Semester: 5th Semester
+
+Course/Program Name: B.Tech Computer Science
+Project Description: A web-based system for managing student course registrations, allowing students to enroll in subjects and view their semester schedule
.
PS C:\Users\DELL\Desktop\Bhoomika-GLA-CourseRegistration> git add course_info.txt
warning: in the working copy of 'course_info.txt', LF will be replaced by CRLF the next time Git touches it
PS C:\Users\DELL\Desktop\Bhoomika-GLA-CourseRegistration> git commit -m "Updated course_info.txt with course/program name and project description"
[master b042622] Updated course_info.txt with course/program name and project description
1 file changed, 3 insertions(+)
PS C:\Users\DELL\Desktop\Bhoomika-GLA-CourseRegistration> git branch course-search
PS C:\Users\DELL\Desktop\Bhoomika-GLA-CourseRegistration> git checkout course-search
Switched to branch 'course-search'
PS C:\Users\DELL\Desktop\Bhoomika-GLA-CourseRegistration> Add-Content src\course_search.txt "Course Search Module'nThis module allows students to search for
available courses by name, subject, or semester. It provides filtering options and displays course details such as course code, credits, and instructor nam
e."
PS C:\Users\DELL\Desktop\Bhoomika-GLA-CourseRegistration> cat src\course_search.txt
Course Search Module
This module allows students to search for available courses by name, subject, or semester. It provides filtering options and displays course details such as
course code, credits, and instructor name.
PS C:\Users\DELL\Desktop\Bhoomika-GLA-CourseRegistration> git add src\course_search.txt
warning: in the working copy of 'src/course_search.txt', LF will be replaced by CRLF the next time Git touches it
PS C:\Users\DELL\Desktop\Bhoomika-GLA-CourseRegistration> git commit -m "Added course search module description"
[course-search 68bc0ce] Added course search module description
1 file changed, 2 insertions(+)
create mode 100644 src/course_search.txt
PS C:\Users\DELL\Desktop\Bhoomika-GLA-CourseRegistration> git branch
* course-search
master
PS C:\Users\DELL\Desktop\Bhoomika-GLA-CourseRegistration> git branch --show-current
course-search
PS C:\Users\DELL\Desktop\Bhoomika-GLA-CourseRegistration> git checkout master
Switched to branch 'master'
PS C:\Users\DELL\Desktop\Bhoomika-GLA-CourseRegistration> ls src

```

Listing branches, current branch, and switching back to master:

```

ls : Cannot find path 'C:\Users\DELL\Desktop\Bhoomika-GLA-CourseRegistration\src' because it does not exist.
At line:1 char:1
+ ls src
+ ~~~~~

+ CategoryInfo          : ObjectNotFound: (C:\Users\DELL\Desktop\Bhoomika-GLA-CourseRegistration\src:String) [Get-ChildItem], ItemNotFoundException
+ FullyQualifiedErrorId : PathNotFound,Microsoft.PowerShell.Commands.GetChildItemCommand

PS C:\Users\DELL\Desktop\Bhoomika-GLA-CourseRegistration> git log --all --oneline --graph
* 68bc0ce (course-search) Added course search module description
* b042622 (HEAD -> master) Updated course_info.txt with course/program name and project description
* 48ccae8 Initial commit: project structure and course_info.txt
PS C:\Users\DELL\Desktop\Bhoomika-GLA-CourseRegistration> git branch --show-current
master
PS C:\Users\DELL\Desktop\Bhoomika-GLA-CourseRegistration> git merge course-search
Updating b042622..68bc0ce
Fast-forward
 src/course_search.txt | 2 ++
1 file changed, 2 insertions(+)
create mode 100644 src/course_search.txt
PS C:\Users\DELL\Desktop\Bhoomika-GLA-CourseRegistration> cat src/course_search.txt
Course Search Module
This module allows students to search for available courses by name, subject, or semester. It provides filtering options and displays course details such as
course code, credits, and instructor name.
PS C:\Users\DELL\Desktop\Bhoomika-GLA-CourseRegistration> git log --oneline --graph
* 68bc0ce (HEAD -> master, course-search) Added course search module description
* b042622 Updated course_info.txt with course/program name and project description
* 48ccae8 Initial commit: project structure and course_info.txt
PS C:\Users\DELL\Desktop\Bhoomika-GLA-CourseRegistration> git branch course-update
PS C:\Users\DELL\Desktop\Bhoomika-GLA-CourseRegistration> git checkout course-update
Switched to branch 'course-update'
PS C:\Users\DELL\Desktop\Bhoomika-GLA-CourseRegistration> cat course_info.txt
Student Name: Bhoomika
Roll Number: 2415001879
Subject Name: <your subject>
Project Name: Online Course Registration System

```

Demonstrating independent development (file absent on master):

Semester: 5th Semester

Course/Program Name: B.Tech Computer Science

Project Description: A web-based system for managing student course registrations, allowing students to enroll in subjects and view their semester schedule.

PS C:\Users\DELL\Desktop\Bhoomika-GLA-CourseRegistration> (Get-Content course_info.txt) -replace 'Semester: 5th Semester', 'Semester: 5th Semester (Odd Semester 2026)' | Set-Content course_info.txt

PS C:\Users\DELL\Desktop\Bhoomika-GLA-CourseRegistration> cat course_info.txt

Student Name: Bhoomika

Roll Number: 2415001879

Subject Name: <your subject>

Project Name: OnLine Course Registration System

Semester: 5th Semester (Odd Semester 2026)

Course/Program Name: B.Tech Computer Science

Project Description: A web-based system for managing student course registrations, allowing students to enroll in subjects and view their semester schedule.

PS C:\Users\DELL\Desktop\Bhoomika-GLA-CourseRegistration>

PS C:\Users\DELL\Desktop\Bhoomika-GLA-CourseRegistration>

PS C:\Users\DELL\Desktop\Bhoomika-GLA-CourseRegistration> git add course_info.txt

PS C:\Users\DELL\Desktop\Bhoomika-GLA-CourseRegistration> git commit -m "Updated semester info in course_info.txt on course-update branch"

[course-update ed3f87b] Updated semester info in course_info.txt on course-update branch

1 file changed, 2 insertions(+), 2 deletions(-)

PS C:\Users\DELL\Desktop\Bhoomika-GLA-CourseRegistration> git checkout master

Switched to branch 'master'

PS C:\Users\DELL\Desktop\Bhoomika-GLA-CourseRegistration> (Get-Content course_info.txt) -replace 'Semester: 5th Semester', 'Semester: V (5th Sem, 2026-27)'

| Set-Content course_info.txt

PS C:\Users\DELL\Desktop\Bhoomika-GLA-CourseRegistration> cat course_info.txt

Student Name: Bhoomika

Roll Number: 2415001879

Subject Name: <your subject>

Project Name: OnLine Course Registration System

Semester: V (5th Sem, 2026-27)

Course/Program Name: B.Tech Computer Science

Project Description: A web-based system for managing student course registrations, allowing students to enroll in subjects and view their semester schedule.

PS C:\Users\DELL\Desktop\Bhoomika-GLA-CourseRegistration> git add course_info.txt

PS C:\Users\DELL\Desktop\Bhoomika-GLA-CourseRegistration> git commit -m "Updated semester format in course_info.txt on master branch"

[master af2b436] Updated semester format in course_info.txt on master branch

1 file changed, 2 insertions(+), 2 deletions(-)

PS C:\Users\DELL\Desktop\Bhoomika-GLA-CourseRegistration> git merge course-update

Auto-merging course_info.txt

CONFLICT (content): Merge conflict in course_info.txt

Question 4 — Merging and Conflict Resolution (8 Marks)

Part A — Merge Feature: Merged course-search into master, verified course_search.txt is available, and displayed Git history.

Fast-forward merge of course-search into master, verification, and log:

```
Semester: 5th Semester

Course/Program Name: B.Tech Computer Science
Project Description: A web-based system for managing student course registrations, allowing students to enroll in subjects and view their semester schedule.
PS C:\Users\DELL\Desktop\Bhoomika-GLA-CourseRegistration> (Get-Content course_info.txt) -replace 'Semester: 5th Semester', 'Semester: 5th Semester (Odd Semester 2026)' | Set-Content course_info.txt
PS C:\Users\DELL\Desktop\Bhoomika-GLA-CourseRegistration> cat course_info.txt
Student Name: Bhoomika
Roll Number: 2415001879
Subject Name: <your subject>
Project Name: Online Course Registration System
Semester: 5th Semester (Odd Semester 2026)

Course/Program Name: B.Tech Computer Science
Project Description: A web-based system for managing student course registrations, allowing students to enroll in subjects and view their semester schedule.
PS C:\Users\DELL\Desktop\Bhoomika-GLA-CourseRegistration>
PS C:\Users\DELL\Desktop\Bhoomika-GLA-CourseRegistration> git add course_info.txt
PS C:\Users\DELL\Desktop\Bhoomika-GLA-CourseRegistration> git commit -m "Updated semester info in course_info.txt on course-update branch"
[course-update ed3f87b] Updated semester info in course_info.txt on course-update branch
1 file changed, 2 insertions(+), 2 deletions(-)
PS C:\Users\DELL\Desktop\Bhoomika-GLA-CourseRegistration> git checkout master
Switched to branch 'master'
PS C:\Users\DELL\Desktop\Bhoomika-GLA-CourseRegistration> (Get-Content course_info.txt) -replace 'Semester: 5th Semester', 'Semester: V (5th Sem, 2026-27)' | Set-Content course_info.txt
PS C:\Users\DELL\Desktop\Bhoomika-GLA-CourseRegistration> cat course_info.txt
Student Name: Bhoomika
Roll Number: 2415001879
Subject Name: <your subject>
Project Name: Online Course Registration System
Semester: V (5th Sem, 2026-27)

Course/Program Name: B.Tech Computer Science
Project Description: A web-based system for managing student course registrations, allowing students to enroll in subjects and view their semester schedule.
PS C:\Users\DELL\Desktop\Bhoomika-GLA-CourseRegistration> git add course_info.txt
PS C:\Users\DELL\Desktop\Bhoomika-GLA-CourseRegistration> git commit -m "Updated semester format in course_info.txt on master branch"
[master af2b436] Updated semester format in course_info.txt on master branch
1 file changed, 2 insertions(+), 2 deletions(-)
PS C:\Users\DELL\Desktop\Bhoomika-GLA-CourseRegistration> git merge course-update
Auto-merging course_info.txt
CONFLICT (content): Merge conflict in course_info.txt
```

Part B — Generate Conflict: Created course-update branch, modified the same line of course_info.txt differently on both branches, attempted merge, hit a conflict, manually resolved it, and committed the resolution.

Creating course-update branch and modifying course_info.txt:

```
Semester: 5th Semester

Course/Program Name: B.Tech Computer Science
Project Description: A web-based system for managing student course registrations, allowing students to enroll in subjects and view their semester schedule.
PS C:\Users\DELL\Desktop\Bhoomika-GLA-CourseRegistration> (Get-Content course_info.txt) -replace 'Semester: 5th Semester', 'Semester: 5th Semester (Odd Semester 2026)' | Set-Content course_info.txt
PS C:\Users\DELL\Desktop\Bhoomika-GLA-CourseRegistration> cat course_info.txt
Student Name: Bhoomika
Roll Number: 2415001879
Subject Name: <your subject>
Project Name: Online Course Registration System
Semester: 5th Semester (Odd Semester 2026)

Course/Program Name: B.Tech Computer Science
Project Description: A web-based system for managing student course registrations, allowing students to enroll in subjects and view their semester schedule.
PS C:\Users\DELL\Desktop\Bhoomika-GLA-CourseRegistration>
PS C:\Users\DELL\Desktop\Bhoomika-GLA-CourseRegistration> git add course_info.txt
PS C:\Users\DELL\Desktop\Bhoomika-GLA-CourseRegistration> git commit -m "Updated semester info in course_info.txt on course-update branch"
[course-update ed3f87b] Updated semester info in course_info.txt on course-update branch
1 file changed, 2 insertions(+), 2 deletions(-)
PS C:\Users\DELL\Desktop\Bhoomika-GLA-CourseRegistration> git checkout master
Switched to branch 'master'
PS C:\Users\DELL\Desktop\Bhoomika-GLA-CourseRegistration> (Get-Content course_info.txt) -replace 'Semester: 5th Semester', 'Semester: V (5th Sem, 2026-27)' | Set-Content course_info.txt
PS C:\Users\DELL\Desktop\Bhoomika-GLA-CourseRegistration> cat course_info.txt
Student Name: Bhoomika
Roll Number: 2415001879
Subject Name: <your subject>
Project Name: Online Course Registration System
Semester: V (5th Sem, 2026-27)

Course/Program Name: B.Tech Computer Science
Project Description: A web-based system for managing student course registrations, allowing students to enroll in subjects and view their semester schedule.
PS C:\Users\DELL\Desktop\Bhoomika-GLA-CourseRegistration> git add course_info.txt
PS C:\Users\DELL\Desktop\Bhoomika-GLA-CourseRegistration> git commit -m "Updated semester format in course_info.txt on master branch"
[master af2b436] Updated semester format in course_info.txt on master branch
1 file changed, 2 insertions(+), 2 deletions(-)
PS C:\Users\DELL\Desktop\Bhoomika-GLA-CourseRegistration> git merge course-update
Auto-merging course_info.txt
CONFLICT (content): Merge conflict in course_info.txt
```

Modifying Semester line differently on each branch and committing both:


```
Automatic merge failed; fix conflicts and then commit the result.
PS C:\Users\DELL\Desktop\Bhoomika-GLA-CourseRegistration> ^C
PS C:\Users\DELL\Desktop\Bhoomika-GLA-CourseRegistration> cat course_info.txt
Student Name: Bhoomika
Roll Number: 2415001879
Subject Name: <your subject>
Project Name: Online Course Registration System
<<<<<< HEAD
Semester: V (5th Sem, 2026-27)
=====
Semester: 5th Semester (Odd Semester 2026)
>>>>>> course-update

Course/Program Name: B.Tech Computer Science
Project Description: A web-based system for managing student course registrations, allowing students to enroll in subjects and view their semester schedule.
PS C:\Users\DELL\Desktop\Bhoomika-GLA-CourseRegistration> (Get-Content course_info.txt) -replace '<<<<<< HEAD', '' -replace 'Semester: V \\\(5th Sem, 2026-27 \\\)', 'Semester: V / 5th Semester (Odd Semester 2026-27)' -replace '=====', '' -replace 'Semester: 5th Semester \\\(Odd Semester 2026\\\\)', '' -replace '>>>>>>'
> course-update', '' | Where-Object { $_.Trim() -ne '' -or $_.eq '' } | Set-Content course_info.txt
PS C:\Users\DELL\Desktop\Bhoomika-GLA-CourseRegistration> notepad course_info.txt
PS C:\Users\DELL\Desktop\Bhoomika-GLA-CourseRegistration> cat course_info.txt
Student Name: Bhoomika
Roll Number: 2415001879
Subject Name: <your subject>
Project Name: Online Course Registration System

Semester: V / 5th Semester (Odd Semester 2026-27)

Course/Program Name: B.Tech Computer Science
Project Description: A web-based system for managing student course registrations, allowing students to enroll in subjects and view their semester schedule.
PS C:\Users\DELL\Desktop\Bhoomika-GLA-CourseRegistration> git add course_info.txt
PS C:\Users\DELL\Desktop\Bhoomika-GLA-CourseRegistration> git commit -m "Resolved merge conflict in course_info.txt between master and course-update"
[master 178b800] Resolved merge conflict in course_info.txt between master and course-update
PS C:\Users\DELL\Desktop\Bhoomika-GLA-CourseRegistration> git status
On branch master
Untracked files:
  (use "git add <file>..." to include in what will be committed)
        cloud_assignment.txt
```

Viewing conflict markers, manual resolution, staging, and commit:

```
PS C:\Users\DELL\Desktop\Bhoomika-GLA-CourseRegistration> cd C:\Users\DELL\Desktop\Bhoomika-GLA-CourseRegistration
>> tree
Folder PATH listing for volume OS
Volume serial number is 00000060 F08F:9F96
C:.
├── backup
├── docs
└── src
PS C:\Users\DELL\Desktop\Bhoomika-GLA-CourseRegistration> git status
On branch master
Untracked files:
  (use "git add <file>..." to include in what will be committed)
        cloud_assignment.txt

nothing added to commit but untracked files present (use "git add" to track)
PS C:\Users\DELL\Desktop\Bhoomika-GLA-CourseRegistration> git log --oneline --graph --all
* 178b800 (HEAD -> master) Resolved merge conflict in course_info.txt between master and course-update
|
| * ed3f87b (course-update) Updated semester info in course_info.txt on course-update branch
| * | af2b436 Updated semester format in course_info.txt on master branch
|/
* 68bc0ce (course-search) Added course search module description
* b042622 Updated course_info.txt with course/program name and project description
* 48ccae8 Initial commit: project structure and course_info.txt
PS C:\Users\DELL\Desktop\Bhoomika-GLA-CourseRegistration> git branch
  course-search
  course-update
* master
PS C:\Users\DELL\Desktop\Bhoomika-GLA-CourseRegistration> git log --oneline --graph
* 178b800 (HEAD -> master) Resolved merge conflict in course_info.txt between master and course-update
|
| * ed3f87b (course-update) Updated semester info in course_info.txt on course-update branch
| * | af2b436 Updated semester format in course_info.txt on master branch
|/
* 68bc0ce (course-search) Added course search module description
* b042622 Updated course_info.txt with course/program name and project description
* 48ccae8 Initial commit: project structure and course_info.txt
PS C:\Users\DELL\Desktop\Bhoomika-GLA-CourseRegistration> git show 178b800
commit 178b80026619c07180bc5f59d7a54df1cc89729e (HEAD -> master)
Merge: af2b436 ed3f87b
```

Question 5 — Final Demonstration (4 Marks)

Final demonstration of: (1) complete Linux project structure, (2) Git repository status and history, (3) feature branches, (4) successful feature merge, and (5) conflict generation and resolution, all shown together.

Project structure, repository status, full history, and branch list:


```

Merge: af2b436 ed3f87b
Author: Bhoomika <your_email@example.com>
Date: Sun Sep 6 00:26:57 2026 +0530

    Resolved merge conflict in course_info.txt between master and course-update

diff --cc course_info.txt
index bd4cb8c,b69abc6..5a96d83
--- a/course_info.txt
+++ b/course_info.txt
@@@ -2,7 -2,7 +2,11 @@@ Student Name: Bhoomik
    Roll Number: 2415001879
    Subject Name: <your subject>
    Project Name: Online Course Registration System
- Semester: V (5th Sem, 2026-27)
- Semester: 5th Semester (Odd Semester 2026)
++
++Semester: V / 5th Semester (Odd Semester 2026-27)
++
++
++

    Course/Program Name: B.Tech Computer Science
    :
commit 178b80026619c07180bc5f59d7a54df1cc89729e (HEAD -> master)
Merge: af2b436 ed3f87b
Author: Bhoomika <your_email@example.com>
Date: Sun Sep 6 00:26:57 2026 +0530

    Resolved merge conflict in course_info.txt between master and course-update

diff --cc course_info.txt
index bd4cb8c,b69abc6..5a96d83
--- a/course_info.txt
+++ b/course_info.txt
@@@ -2,7 -2,7 +2,11 @@@ Student Name: Bhoomik
    Roll Number: 2415001879
    Subject Name: <your subject>
    Project Name: Online Course Registration System
- Semester: V (5th Sem, 2026-27)
- Semester: 5th Semester (Odd Semester 2026)

```

Merge commit details confirming conflict resolution (git show):

